Culturing microorganisms

Grow & calculate

1 Binary fission



One bacterium → two

One bacterial cell → two. With nutrients and suitable temperature, division may take **20 minutes;** use the given time.

1 → 2 → 4 → 8: repeated doubling, not addition. No nucleus; bacteria do not use mitosis.

2 Repeated doubling

n = total time ÷ division time.
Final population = **start × 2^n.**

50 bacteria; 20-minute divisions; 2 h: 120 ÷ 20 = 6 doublings.

50×2^6 = **3,200 = 3.2×10^3.** Keep time units consistent; apply the stated growth conditions.

Prevent contamination

3 Grow an uncontaminated culture

Broth: nutrients for growth. **Agar:** surface for colonies (a visible group from an initial cell/group).

Uncontaminated cultures allow fair antimicrobial comparisons. **Aseptic technique** keeps unwanted microorganisms out.

Sterilise dishes/medium to remove unwanted microbes. **Flame the loop; cool it** so it does not kill the culture.

Open lid briefly to limit airborne contamination. Follow school handling/disposal procedures.

4 Safe incubation

Tape edges, not a full seal: prevents opening; allows gas exchange.

Invert: condensation cannot drip onto culture. **Maximum 25°C** in school limits harmful human pathogens. Do not open incubated plates.

Compare inhibition

5 Compare antimicrobial discs

Use asepsis to prepare a lawn on sterile agar. Add **antibiotic/antiseptic discs** and a **solvent-only control.**

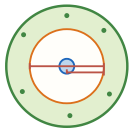
Incubate safely; measure inhibition zones. Repeat and calculate a mean for each substance.

Change: substance/type or concentration. Measure: **zone diameter or area.**

Keep strain, agar, disc size, substance volume, time and temperature constant.

Larger zone → more inhibition **under these conditions.** Different diffusion rates mean this does not prove effectiveness inside a patient.

6 Zone area



Diameter → halve → radius

Radius = diameter ÷ 2; area = πr^2.

Diameter 12 mm → radius 6 mm → $\pi \times 6^2$ = **113 mm^2** (3 s.f.).

Use squared units. Subtract disc area only if asked. A clear zone means **growth was inhibited.**